Position, path and the reference frame

Kinematics begins by saying where, when, and relative to what.

A coordinate is not the motion

A 1-D frame is an origin, a positive direction, and a clock. Position $x(t)$ is measured from that origin. Changing origin changes coordinates, not the physical path.

$$x(t)\ [\mathrm{m}], \qquad t\ [\mathrm{s}]$$

WHY / Write the sign convention before equations. "Left" is not automatically negative; it is negative only if left is chosen as the negative axis.

Displacement is an endpoint difference

From x_i to x_f, displacement is signed. A round trip has zero displacement even though motion occurred. Displacement depends only on endpoints, not the route.

$$\Delta x = x_f - x_i$$

Move from 2 m to 7 m, then to 4 m: net displacement is +2 m.

Distance is accumulated path length

Distance never decreases and never carries a direction. It adds every segment, including retracing. For a differentiable path, add the magnitude of each small displacement.

$$d = \int_{t_i}^{t_f} |v(t)|\, dt \geq |\Delta x|$$

WATCH / Do not replace distance by $|x_f - x_i|$ after a reversal; that misses the path already travelled.

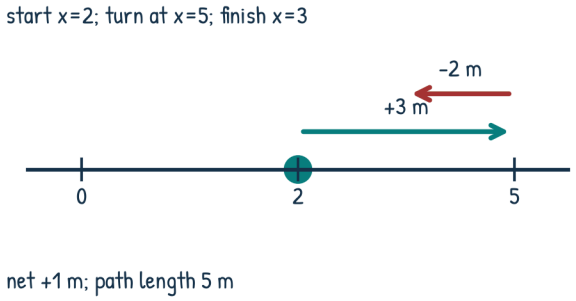
A quick counterexample

A walker goes east 5 m and west 3 m. The final point is 2 m east of the start, but the route length is 8 m. The ratio $|\Delta x|/d$ lies from 0 to 1.

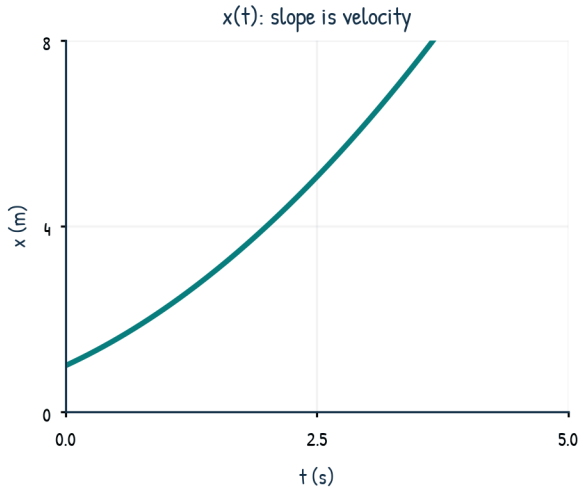
$$\Delta x = +2\ \mathrm{m}, \qquad d = 8\ \mathrm{m}$$

Equality $d = |\Delta x|$ holds only when direction never reverses.

position: physical sketch



position: diagnostic graph



REMEMBER / Choose origin, axis, and clock first; displacement is final minus initial position, while distance counts the whole path.

Average and instantaneous velocity

Velocity is the signed rate of position change; speed is its magnitude.

Average velocity uses net change

Over a time interval, average velocity is displacement divided by elapsed time. It can be positive, negative, or zero; it is not average speed.

$$\bar{v} = \frac{x(t_2) - x(t_1)}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$$

WHY / A round trip has zero average velocity because its net displacement is zero, even though its average speed is positive.

Instantaneous velocity is a limit

Shrink the interval around the instant. The limiting secant slope becomes the tangent slope of x versus t. A positive or negative sign identifies direction on the chosen axis.

$$v(t) = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

If $x = 3t^2 - 2t$, then $v = 6t - 2$; at $t = 1$ s, $v = 4$ m/s.

Speed forgets direction

Speed is the non-negative magnitude of velocity. Average speed uses total distance, so it is at least the magnitude of average velocity.

$$\text{speed} = |v|, \qquad \bar{s} = \frac{d}{\Delta t} \geq |\bar{v}|$$

WATCH / Do not average signed velocities and call the result average speed.

Relative position changes the frame

For two objects measured in the same frame, define $x_{AB} = x_A - x_B$. Its derivative is the velocity of A as seen from B. This definition fixes the subtraction order.

$$x_{AB} = x_A - x_B, \qquad v_{AB} = v_A - v_B$$

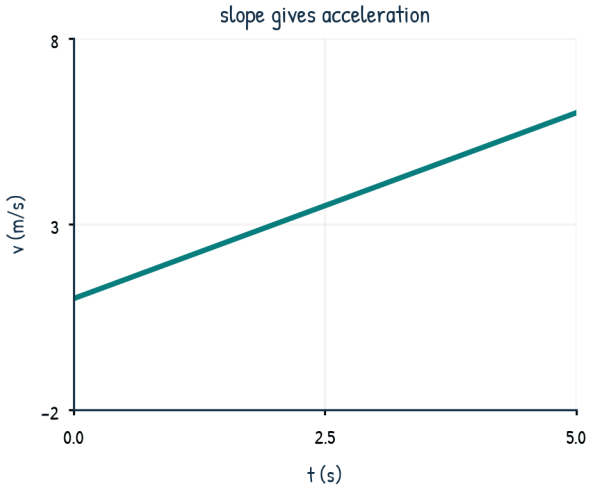
If A moves at 8 m/s right and B at 3 m/s right, A gains on B at 5 m/s.

velocity: physical sketch



same frame; v_A/B = 2 m/s

velocity: diagnostic graph



REMEMBER / A secant slope gives average velocity; a tangent slope gives instantaneous velocity.

Acceleration and the meaning of signs

Acceleration measures velocity change, not "slowing down."

Average and instantaneous acceleration

Acceleration is the rate of velocity change. It may be positive, negative, or zero; the sign describes direction of velocity change, not whether the object is speeding up.

$$\bar{a} = \frac{v_f - v_i}{\Delta t}, \qquad a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

WHY / A car can move right while acceleration points left: it is moving right but losing speed.

Use the sign product

For one-dimensional motion at nonzero speed, the rate of change of speed has the sign of va. Thus speed increases when $va > 0$ and decreases when $va < 0$.

$$\frac{d|v|}{dt} = \operatorname{sgn}(v)\, a \quad (v \neq 0)$$

At a turning point $v = 0$, inspect the motion on each side; the sign test is not defined at that instant.

Four sign combinations

Rightward motion with rightward acceleration speeds up; rightward with leftward acceleration slows. Leftward with leftward acceleration speeds up left; leftward with rightward acceleration slows.

$$(v > 0, a > 0) \text{ or } (v < 0, a < 0): \text{ speed rises}$$

WATCH / "Negative acceleration" is not a synonym for deceleration. The axis orientation may be reversed.

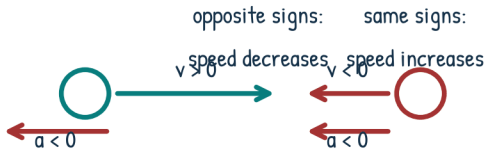
Dimensions expose mistakes

Velocity has dimensions length/time; acceleration has length/time squared. A slope on an x-t graph has velocity units; a slope on a v-t graph has acceleration units.

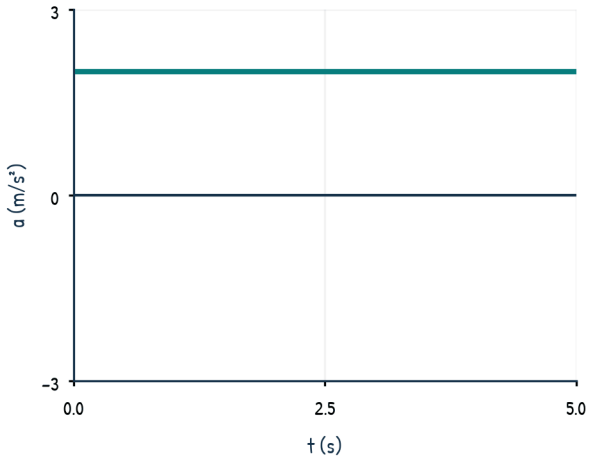
$$[v] = LT^{-1}, \qquad [a] = LT^{-2}$$

A claimed acceleration in m/s is missing one factor of time in the denominator.

accel: physical sketch



accel: diagnostic graph



REMEMBER / Same signs of v and a increase speed; opposite signs decrease speed.

Position, velocity and acceleration graphs

A graph is a calculator: slopes differentiate and signed areas integrate.

Read x versus t

The slope of a chord is average velocity; the tangent slope is instantaneous velocity. A horizontal tangent means $v = 0$ at that instant, not necessarily that the particle stays at rest.

$$\text{slope of } x(t) = \frac{dx}{dt} = v$$

WATCH / A curved x-t graph still describes one-dimensional motion; the curve is not the path through space.

Read v versus t

Slope gives acceleration. Signed area from t_1 to t_2 gives displacement. To get distance, split at every zero of v and add the absolute areas.

$$\text{slope} = a, \qquad \int_{t_1}^{t_2} v\,dt = \Delta x$$

If v changes sign, net displacement can be small while total distance is large.

Read a versus t

The signed area under acceleration gives change in velocity. Add the initial velocity to recover $v(t)$. Acceleration alone does not give position until you integrate again and supply initial data.

$$v(t_2) - v(t_1) = \int_{t_1}^{t_2} a\,dt$$

WHY / The integration constants are physical initial conditions, not optional algebra.

Cross-check slopes and areas

If a is constant, v-t is a straight line and x-t is a parabola. If v is constant, x-t is a straight line and a-t is zero. Use these shape relations to catch sign errors.

$$a = \text{constant} \Rightarrow v(t)\ \text{linear}, \quad x(t)\ \text{quadratic}$$

A positive triangular area below the t-axis is negative displacement, not negative distance.

graphs: physical sketch

x-t: tangent slope = v

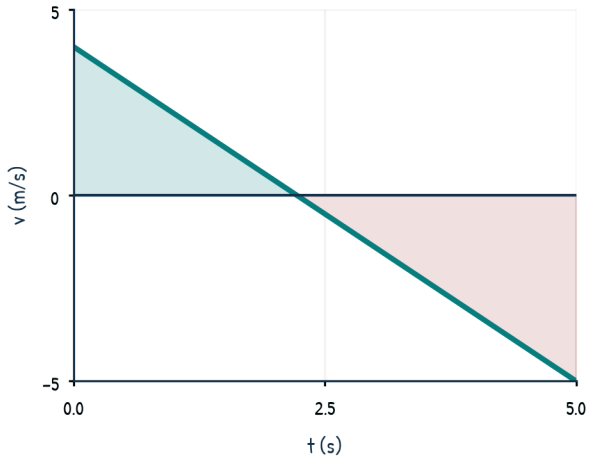
v-t: slope = a; signed area = Δx

a-t: signed area = Δv



integrate downward; differentiate upward

graphs: diagnostic graph



REMEMBER / x-t slope is v; v-t slope is a and signed area is displacement; a-t area is change in v.

Constant-acceleration equations

Derive the familiar equations from integration; use them only when a is constant.

Velocity after time t

For constant acceleration, integrate $dv/dt = a$ and apply $v(0) = u$. The result is linear in time. The sign of a is retained. If v and a share a sign, speed grows; if their signs oppose, speed falls until a turning point.

$$v = u + at$$

At $a = 0$, velocity remains u.

Displacement from the v-t area

The signed trapezoid under a straight v-t line has area average endpoint velocity times time. This gives displacement, including cases where one endpoint velocity is negative; if a reversal lies inside, signed area still works.

$$\Delta x = ut + \tfrac{1}{2}at^2 = \tfrac{u+v}{2}t$$

WHY / The trapezoid form uses endpoint velocities; the polynomial form uses initial velocity and acceleration.

Eliminate time

From $v - u = at$, substitute into displacement or use $a = v\,dv/dx$ with constant a. The resulting relation does not require time, but still assumes constant acceleration.

$$v^2 = u^2 + 2a\Delta x$$

WATCH / A position-dependent or speed-dependent acceleration invalidates this equation unless a is constant over the interval.

Choose the equation by the missing variable

Known u, a, t: find v or displacement. Known u, v, t: use the trapezoid. Known u, v, a: eliminate t. Set a clear signed coordinate before substitution.

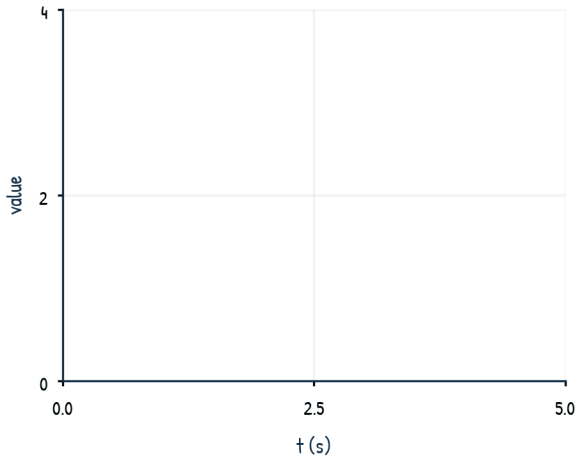
$$\left\{v = u + at, \quad \Delta x = ut + \tfrac{1}{2}at^2, \quad v^2 = u^2 + 2a\Delta x\right\}$$

Every term in an equation for displacement must have units of length.

constant: physical sketch

Kinematics diagram

constant: diagnostic graph



REMEMBER / Every constant-a formula comes from the straight-line v-t graph; time-free form follows by elimination.

Free fall and vertical throws

Near Earth, gravity is approximately constant and points downward.

Choose the axis once

With upward positive and negligible air resistance, acceleration is $-g$. Use the same convention for initial velocity, displacement, and final velocity; do not switch signs mid-solution.

$$a = -g, \qquad g \approx 9.8\ \mathrm{m/s^2}$$

WHY / At the top, v is momentarily zero but gravity still accelerates the ball downward.

Rise from launch level

For initial upward speed $u > 0$, the apex occurs when $v = 0$. The rise and ascent time follow from constant-a kinematics.

$$t_{\mathrm{up}} = \frac{u}{g}, \qquad H = \frac{u^2}{2g}$$

If u doubles, the maximum rise becomes four times as large.

Return to the launch elevation

In vacuum, motion is symmetric about the apex for equal start/end elevations. The total time is twice the ascent time; the return velocity has the same magnitude and opposite sign.

$$T = \frac{2u}{g}, \qquad v(T) = -u$$

WATCH / This symmetry fails with drag or when comparing unequal elevations.

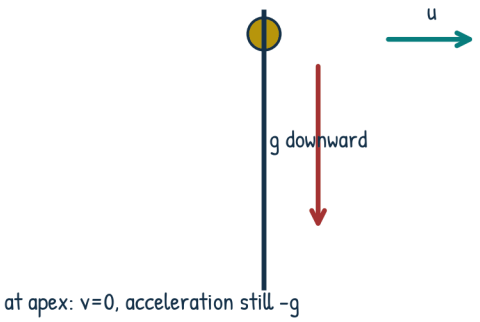
General vertical position equation

For launch position y_0 and signed initial velocity u, solve $y - y_0 = ut - gt^2/2$. A height can be crossed twice; retain non-negative roots and interpret direction from v.

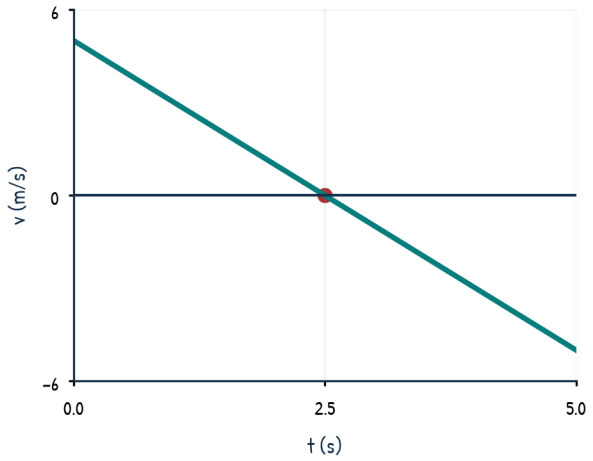
$$y(t) = y_0 + ut - \tfrac{1}{2}gt^2, \qquad v(t) = u - gt$$

For a drop from rest through height h: $t = \sqrt{2h/g}$ and impact speed $\sqrt{2gh}$.

freefall: physical sketch



freefall: diagnostic graph



REMEMBER / Choose up positive: a = -g at every point, including the apex.

Displacement during the nth second

An interval formula gives signed displacement, not automatically distance.

Derive by subtraction

For constant a, write $x(t) - x(0) = ut + at^2/2$. Subtract the displacement at $n-1$ seconds from that at n seconds.

$$\Delta x_n = x(n) - x(n-1) = u + \tfrac{a}{2}(2n-1)$$

WHY / Here n counts one-second intervals. For intervals of duration delta, subtract x(ndelta)-x((n-1)delta) instead.

Signed interval result

The expression can be negative if the particle moves left overall during that second. It is the area under v-t between n-1 and n seconds.

$$\Delta x_n = \int_{n-1}^{n} v(t)\,dt$$

With u=0, a=4 m/s², the third-second displacement is 10 m.

When is it a distance?

If v keeps one sign throughout the interval, distance is $|\Delta x_n|$. If v crosses zero, split at $t_* = -u/a$ and add the magnitudes of the two signed pieces.

$$d_n = \int_{n-1}^{n} |u + at|\,dt$$

WATCH / Do not call a negative nth-second displacement a negative distance.

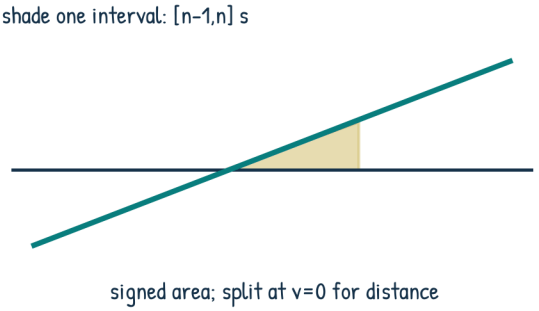
A reversal example

Let $u = 2$ m/s and $a = -4$ m/s². The turn occurs at 0.5 s. During the first second the signed displacement is zero, but the particle travels 0.5 m out and 0.5 m back.

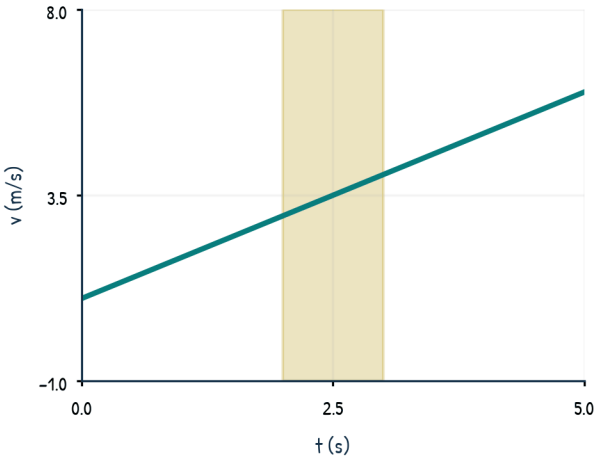
$$\Delta x_1 = 2 - 2 = 0, \qquad d_1 = 1\ \text{m}$$

This is why distance must be found by absolute velocity area.

nth: physical sketch



nth: diagnostic graph



REMEMBER / Subtract x(n) - x(n-1); if velocity reverses within the interval, split and take absolute areas.

Variable acceleration: integrate with initial data

If a changes, return to definitions instead of using constant-a formulas.

Acceleration given as a(t)

Integrate acceleration to obtain velocity, then integrate velocity to obtain position. At each stage insert the corresponding initial condition.

$$v(t) = v_0 + \int_0^t a(\tau)\,d\tau, \qquad x(t) = x_0 + \int_0^t v(\tau)\,d\tau$$

For $a = 6t$, v(0)=2, x(0)=0: $v = 3t^2 + 2$, $x = t^3 + 2t$.

Acceleration given as a(v)

Use $dv/dt = a(v)$ to separate variables and find elapsed time as a function of velocity. The integral must respect the initial and final velocities and any singular points.

$$t - t_0 = \int_{v_0}^{v} \frac{dw}{a(w)}$$

WHY / This route can reveal terminal speed as an equilibrium where a(v)=0.

Acceleration given as a(x)

Chain rule gives $dv/dt = (dv/dx)(dx/dt) = v\,dv/dx$. It converts acceleration-versus-position into an integral for v squared.

$$v\,dv = a(x)\,dx, \qquad \tfrac{1}{2}(v^2 - v_0^2) = \int_{x_0}^{x} a(X)\,dX$$

WATCH / The result fixes v²; choose the velocity sign on each branch and split at a turning point.

The method must match the given variable

Use t when a is specified in time, v when it is specified in speed, x when position dependence is explicit. If acceleration depends on multiple variables, solve the actual differential equation or use a justified substitution.

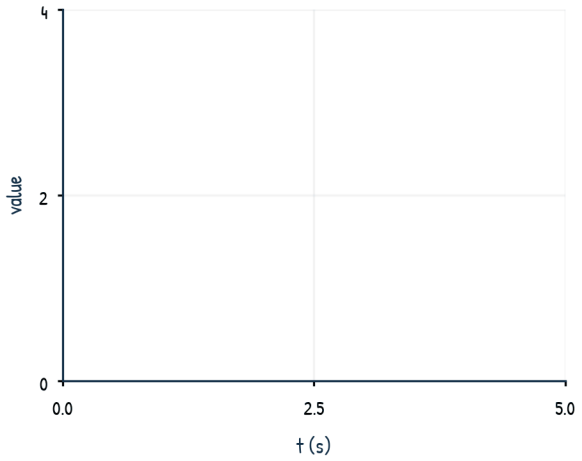
$\text{constant } a:\ \text{algebra}; \quad a(t):\ \text{integrate}; \quad a(x): v\,dv/dx$

Initial conditions are required to recover the physical trajectory.

variable: physical sketch



variable: diagnostic graph



REMEMBER / Given a(t), integrate twice; use initial velocity and position to determine both constants.

The v dv/dx method and turning points

The chain rule converts a second-order time law into an energy-like position relation.

Derive the identity

For a differentiable trajectory, apply the chain rule to v as a function of x along the motion. This does not require constant acceleration.

$$a = \frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}$$

WHY / This is especially useful when acceleration depends on position but time is absent.

Example: restoring acceleration

Suppose $a = -kx$ with k positive and initial state x=0, v=v0. Integrate from 0 to x. The accessible positions satisfy v squared non-negative.

$$v^2 = v_0^2 - kx^2$$

Turning points occur at $x = \pm v_0/\sqrt{k}$. The result describes bounded motion.

Sign branches matter

The equation gives $v = \pm\sqrt{v_0^2 - kx^2}$. Choose the sign from the direction of travel; after a turning point the sign changes while the same v² relation can still hold.

$$v^2 \geq 0 \quad \Longrightarrow \quad |x| \leq v_0/\sqrt{k}$$

WATCH / Do not take the positive square root for both the outward and return branches.

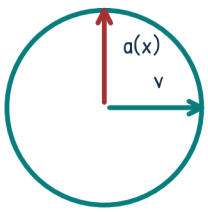
Check the dimensions and limit

For $a = -kx$, k has units s^-2. Then v_0^2/k has units m². As k tends to zero, the turning point recedes without bound, consistent with vanishing restoring acceleration.

$$[k] = T^{-2}, \qquad [v_0^2/k] = L^2$$

At x=0, the formula returns the initial speed magnitude v_0.

vdv: physical sketch

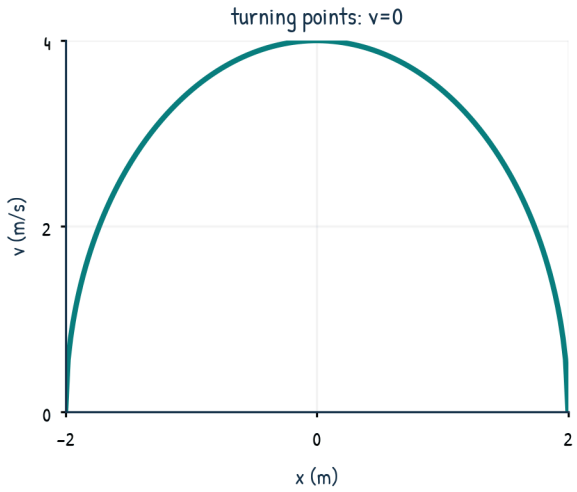


a = v dv/dx

integrate to v²(x)

turn where v² = 0

vdv: diagnostic graph



REMEMBER / Integrate v dv = a(x) dx, keep the integration constant, then check where v² can remain non-negative.

Relative motion on a line

Subtract velocities in one shared frame; use signed separation to decide approach.

Define relative position first

Set $r = x_A - x_B$. Then $dr/dt = v_A - v_B$. If r is positive, A is to the right; the gap shrinks when r and dr/dt have opposite signs.

$$r_{AB} = x_A - x_B, \qquad v_{AB} = v_A - v_B$$

A at x=10 moving left at 2, B at x=0 moving right at 1: r>0 and v_AB=-3, so they approach.

Closing speed is a magnitude

Closing speed is $-|dr/dt|$ only when the pair is approaching; its positive magnitude is $|v_A - v_B|$. Equal direction speeds subtract; opposite directions add.

$$\text{approach if } r\,v_{AB} < 0; \qquad \text{closing speed} = |v_{AB}|$$

WATCH / Relative velocity is signed. "Closing speed" is non-negative and only meaningful while the gap decreases.

Constant relative acceleration

When both accelerations are constant, relative position obeys the same kinematic equations with $a_{AB} = a_A - a_B$. This can reduce two trajectories to one gap equation.

$$a_{AB} = a_A - a_B, \qquad r(t) = r_0 + v_{AB,0}t + \tfrac{1}{2}a_{AB}t^2$$

WHY / A catch or meeting is a root of r(t)=0; check the root is in the physical time interval.

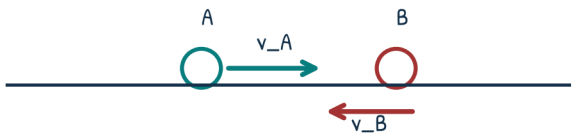
Worked catch-up check

A leads by 100 m. A moves at 10 m/s, B follows at 15 m/s. The gap closes at 5 m/s, so catch-up takes 20 s; both positions then agree.

$$t = \frac{100}{15 - 10} = 20 \text{ s}$$

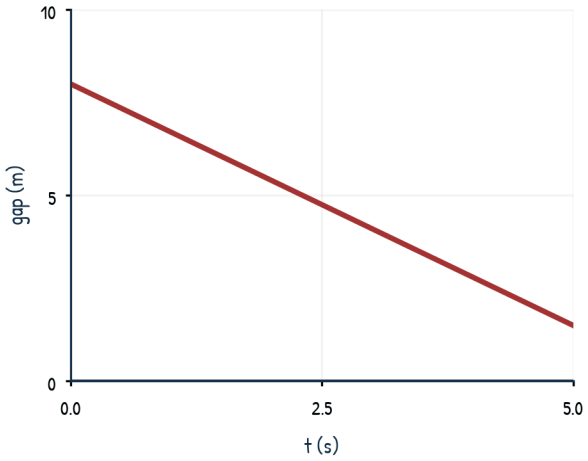
For two trains approaching at 80 and 60 km/h, closing speed is 140 km/h, not 20 km/h.

relative1d: physical sketch



r=x_A-x_B; approach if r dr/dt < 0

relative1d: diagnostic graph



REMEMBER / v_AB = v_A - v_B. The sign of relative velocity alone is not the closing direction; check separation too.

Piecewise motion and stopping distance

Break a motion at every change in acceleration or constraint; match state between phases.

Build a state ledger

For each phase record start time, duration, initial x and v, acceleration, final x and v. A phase ends with the next phase's initial state unless an impact or impulse changes velocity.

$$(x_i, v_i) \xrightarrow{a_i,\ \Delta t_i} (x_f, v_f)$$

WHY / The ledger prevents accidental resets of position or velocity when a changes.

Brake under constant deceleration

For initial speed u and deceleration magnitude b, set signed acceleration opposite the motion. Solve $0 = u^2 - 2bd$ for stopping distance, or $0 = u - bt$ for stopping time.

$$t_{\text{stop}} = \frac{u}{b}, \qquad d_{\text{brake}} = \frac{u^2}{2b}$$

Doubling u doubles stopping time but quadruples braking distance if b is unchanged.

Reaction distance is separate

During a reaction delay t_r, the driver has not yet braked; at unchanged speed the extra distance is ut_r. Total stopping distance is reaction distance plus braking distance.

$$d_{\text{stop}} = ut_r + \frac{u^2}{2b}$$

WATCH / Do not call the braking distance the total stopping distance when reaction time is supplied.

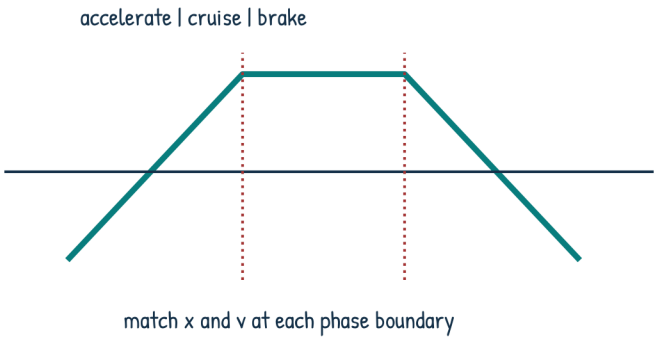
Piecewise graph audit

On a v-t graph, each phase is a segment. Add signed areas for total displacement; add absolute areas after splitting at every zero of v for distance.

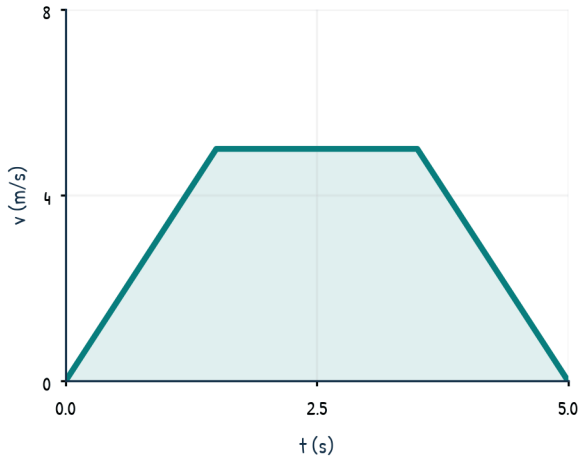
$$\Delta x = \sum_i \int_i v\,dt, \qquad d = \sum_i \int_i |v|\,dt$$

If a phase starts with negative velocity, positive acceleration may first reduce speed before reversing direction.

piecewise: physical sketch



piecewise: diagnostic graph



REMEMBER / Position and velocity carry across a smooth phase boundary; braking distance scales as speed squared.

Projectile motion starts with components

Resolve launch velocity; gravity changes only the vertical component in the vacuum model.

Choose axes and initial components

Take x horizontal and y upward. A launch speed v0 at angle theta above horizontal has components found by projection onto the axes.

$$v_{0x} = v_0 \cos\theta, \qquad v_{0y} = v_0 \sin\theta$$

At theta=0 the launch is horizontal; at theta=90 degrees it is vertical.

Independent accelerations

Neglecting drag, the only acceleration is gravity. Horizontal velocity is constant; vertical velocity decreases linearly.

$$a_x = 0, \quad a_y = -g; \qquad v_x = v_0 \cos\theta, \quad v_y = v_0 \sin\theta - gt$$

WHY / A horizontal launch and a vertical drop from the same height have identical vertical motion.

Parametric trajectory

With launch point at the origin, integrate each component. The equations hold until the projectile hits a surface or the assumptions fail.

$$x = v_0 \cos\theta\, t, \quad y = v_0 \sin\theta\, t - \tfrac{1}{2} g t^2$$

WATCH / The path in space is not the x-t graph. x(t) and y(t) are coordinates at the same time.

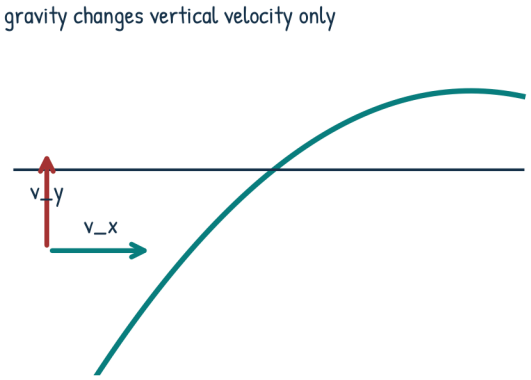
Independence test

At any fixed time, vertical drop is $gt^2/2$, independent of horizontal launch speed. In the same ideal conditions, horizontal and dropped balls hit together.

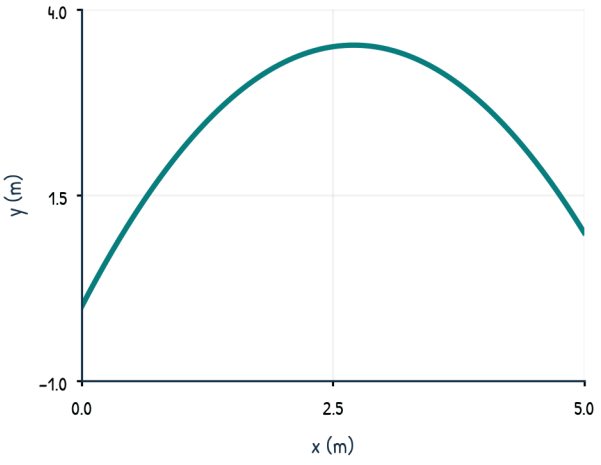
$$\Delta y_{\text{fall}} = \tfrac{1}{2} g t^2$$

Changing horizontal speed changes where the ball lands, not when it reaches a given vertical level.

projectile: physical sketch



projectile: diagnostic graph



REMEMBER / Horizontal and vertical equations share time t; their only coupling is the common clock.

Horizontal launch from a height

The vertical fall determines flight time; horizontal speed sets range.

Solve time from vertical motion

A horizontal launch has initial vertical velocity zero. With the ground h below the launch point, solve the vertical displacement equation for the positive flight time.

$$-h = -\tfrac{1}{2}gt^2 \Rightarrow t = \sqrt{\tfrac{2h}{g}}$$

Time is independent of horizontal speed in the no-drag model.

Find horizontal range

Horizontal acceleration is zero, so the range from the cliff edge is constant horizontal speed times flight time. A vertical launch has zero range but still takes finite time to fall from a nonzero height.

$$R = v_0 t = v_0\sqrt{\tfrac{2h}{g}}$$

WHY / The equation assumes a level landing surface at height h below launch.

Impact velocity and speed

At impact, horizontal velocity remains v0 and vertical velocity is downward with magnitude gt. Use vector components for direction and Pythagoras for speed.

$$v_x = v_0, \quad v_y = -\sqrt{2gh}, \quad |\mathbf{v}| = \sqrt{v_0^2 + 2gh}$$

WATCH / Impact speed is not the horizontal speed; gravity has added vertical kinetic motion.

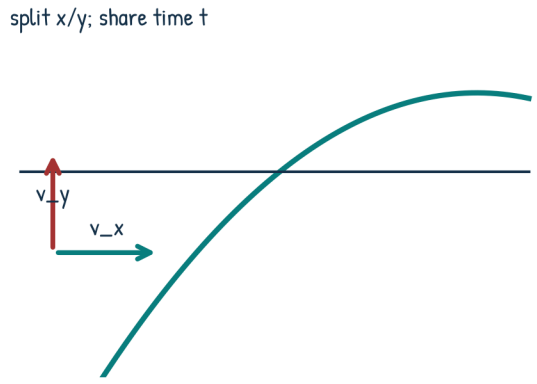
Worked scale check

For v0=15 m/s, h=20 m, g=10 m/s², flight time is 2 s and range is 30 m. The vertical impact speed is 20 m/s.

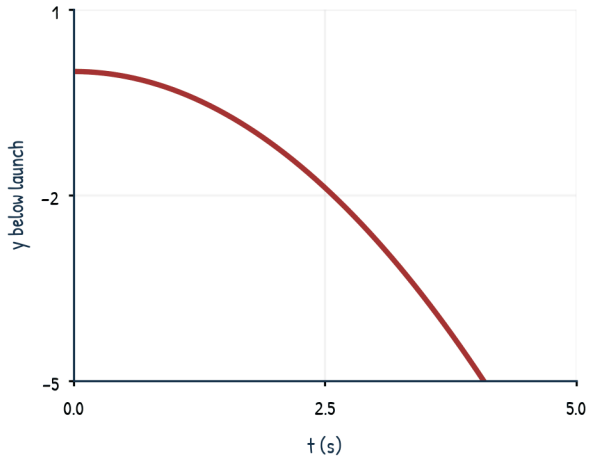
$$t = 2\ \text{s}, \quad R = 30\ \text{m}, \quad v_{\text{impact}} = 25\ \text{m/s}$$

The 25 m/s impact speed follows from sqrt(15²+20²), a useful independent check. The 30 m range is horizontal displacement only; the actual curved path is longer.

horizontal: physical sketch



horizontal: diagnostic graph



REMEMBER / For a level ground h below launch: t = sqrt(2h/g), R = v0 sqrt(2h/g).

Oblique launch: time, height and range

For launch and landing at equal elevations, solve y(T)=0 for the nonzero root.

Time to return to launch elevation

Set vertical displacement to zero. The root t=0 is launch; the second root is the return time. It is positive for an upward launch; a horizontal launch from the same level has no later return.

$$0 = v_0 \sin\theta\, T - \tfrac{1}{2} g T^2 \Rightarrow T = \frac{2 v_0 \sin\theta}{g}$$

WATCH / This is not the flight time to a cliff floor or a higher platform.

Maximum height

At the apex the vertical component vanishes. Apply the vertical constant-a equation to obtain height above launch point. At fixed launch speed, a larger vertical component raises the apex but leaves less speed for horizontal motion.

$$H = \frac{v_0^2 \sin^2\theta}{2g}$$

Only vertical launch speed contributes to the height.

Horizontal range

The horizontal component is constant. Multiply it by flight time and use the double-angle identity. The result is a signed x displacement; if the chosen launch direction points backward, keep the negative sign.

$$R = v_0 \cos\theta\, T = \frac{v_0^2 \sin 2\theta}{g}$$

WHY / Range is maximized at 45 degrees only when launch and landing elevations match and speed is fixed.

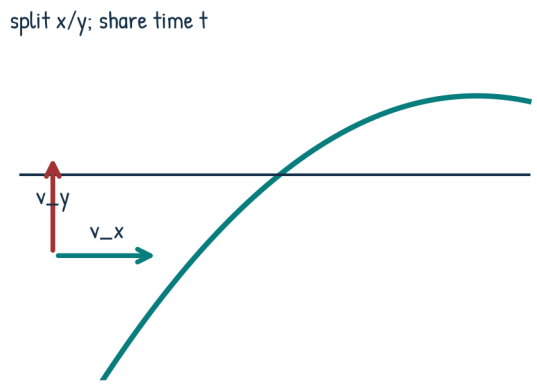
Useful elimination

For the equal-height case, combine range and height to relate trajectory shape without v0 or g. At 30 degrees, R/H is 4 times sqrt(3); the ratio compares horizontal reach with peak rise.

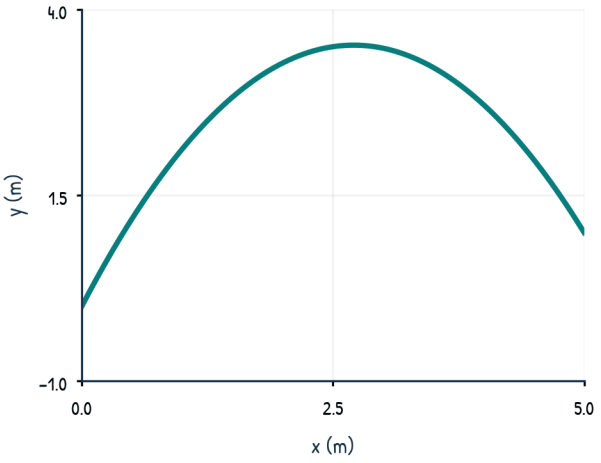
$$\frac{R}{H} = 4\cot\theta$$

At a vertical launch, R=0; at 45 degrees, R is maximum.

projectile general: physical sketch



projectile general: diagnostic graph



REMEMBER / T = 2v0 sintheta/g, H = v0² sin²theta/(2g), R = v0² sin2theta/g under equal-height vacuum conditions.

Trajectory, velocity and speed at a height

Eliminate time for the path; retain components for velocity and angle.

Eliminate time to get y(x)

From x=v0 costheta t, solve for t and substitute into y(t). This assumes costheta is nonzero; vertical launch is a separate straight-line limit.

$$y = x\tan\theta - \frac{gx^2}{2v_0^2\cos^2\theta}$$

The quadratic coefficient is negative for g>0, so the path curves downward.

Velocity direction along the path

Horizontal velocity stays fixed; vertical velocity changes with time. The flight-path angle phi satisfies tanphi=vy/vx whenever vx is nonzero.

$$\tan\phi = \frac{v_0\sin\theta - gt}{v_0\cos\theta}$$

WHY / At the apex phi=0; the velocity is horizontal, not zero unless the launch was vertical.

Speed at vertical rise y

With gravity only, $v_x^2 + v_y^2 = v_0^2 - 2gy$ for launch from y=0. It is an energy-like kinematic identity; it does not determine direction by itself.

$$v^2 = v_0^2 - 2gy$$

WATCH / If y exceeds v0²/(2g), the right side is negative: that height is unreachable.

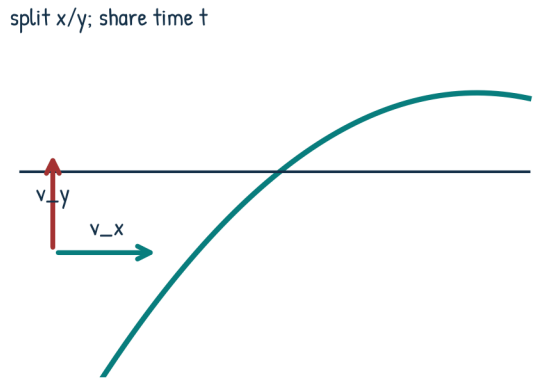
Same height, same speed

For a vacuum projectile, at any later point at the launch elevation the speed magnitude returns to v0. At a different height, vertical speed magnitude is fixed by the equation above.

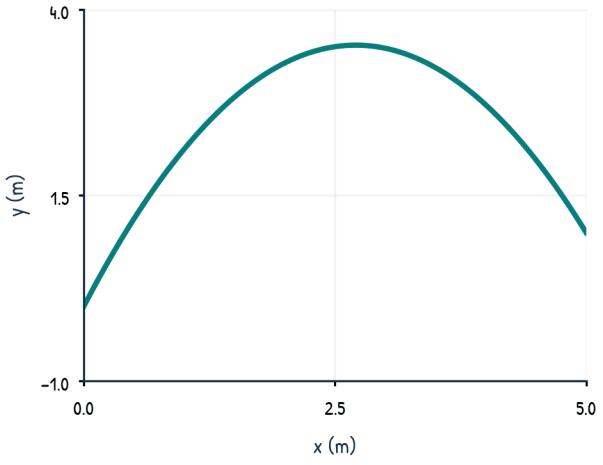
$$v(y=0) = v_0$$

For ground-to-ground flight, the landing angle is -theta and the landing speed equals launch speed.

trajectory: physical sketch



trajectory: diagnostic graph



REMEMBER / The vacuum trajectory is a downward-opening parabola; at a given height the speed is launch speed adjusted by vertical rise.

Range twins and the maximum range

The sine of twice the angle creates complementary-angle symmetry.

Complementary angles are range twins

For equal launch and landing elevations, R depends on sin 2theta. Replacing theta by 90°-theta leaves this sine unchanged.

$$R(\theta) = R(90^\circ - \theta)$$

A 30-degree shot and a 60-degree shot have equal range at the same speed.

But their arcs differ

The steeper shot has larger vertical component, so it reaches greater height and stays in flight longer, while the horizontal component is smaller.

$$T \propto \sin\theta, \qquad H \propto \sin^2\theta$$

WHY / Equal range does not imply equal trajectory, impact time, or impact velocity vector.

Maximum range condition

At fixed v0 and g, maximize sin 2theta over 0 to 90 degrees. The maximum is one at theta=45 degrees; the two angle solutions for any shorter equal-height range merge here.

$$R_{\max} = \frac{v_0^2}{g} \quad (\theta = 45^\circ)$$

WATCH / A different target height, wind, drag law, or launch constraint changes the optimum.

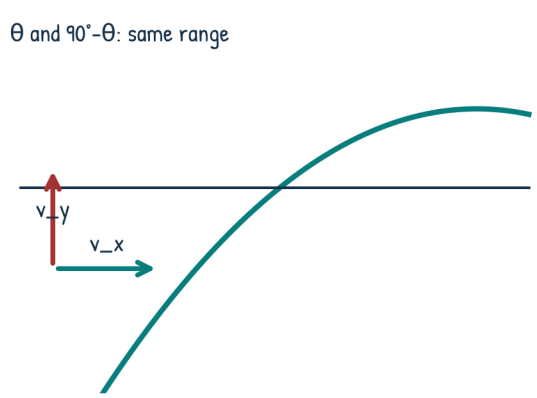
Two angles for a target

For horizontal ground at distance R below the maximum, solve sin 2theta = Rg/v0². If 0<R<Rmax, there are generally two launch angles; at Rmax they merge.

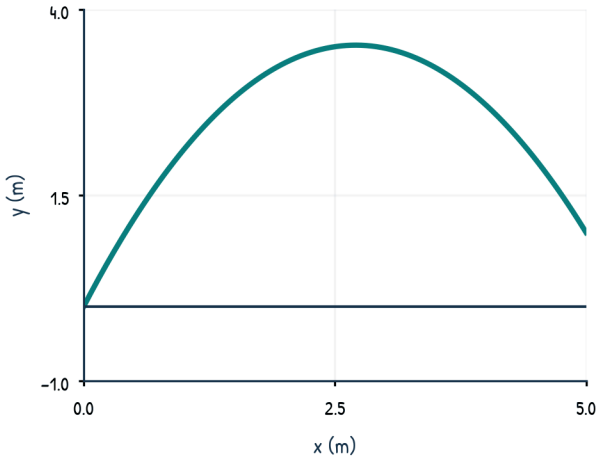
$$\sin 2\theta = \frac{Rg}{v_0^2}$$

No equal-height launch angle reaches a target beyond Rmax in the no-drag model.

range: physical sketch



range: diagnostic graph



REMEMBER / Angles theta and 90°-theta give the same range, but not the same height or flight time.

Launch and landing elevations differ

Use the full vertical quadratic; the range formula no longer applies.

Set the vertical target condition

Let launch height be y0 and target height be yf. Solve $yf - y0 = v0sinthetat - gtfff1/2$. Keep the positive root that corresponds to the physical landing event.

$$\tfrac{1}{2}gt^2 - v_0\sin\theta\, t + (y_f - y_0) = 0$$

WATCH / A negative or pre-launch root is not a flight time. A higher platform may have two crossings.

Range to the target level

Once T is known, horizontal range from launch is $v0costhetaT$. If the projectile is launched from a height h above a level floor, use yf-y0=-h.

$$R = v_0\cos\theta\, T$$

For theta=0 and h>0, this reduces to the horizontal-launch result.

Height crossing can happen twice

A target elevation below the apex can be reached once on ascent and once on descent. The two roots have vertical velocities of opposite sign.

$$v_y(T) = v_0\sin\theta - gT$$

WHY / Root signs tell which branch intersects the specified height.

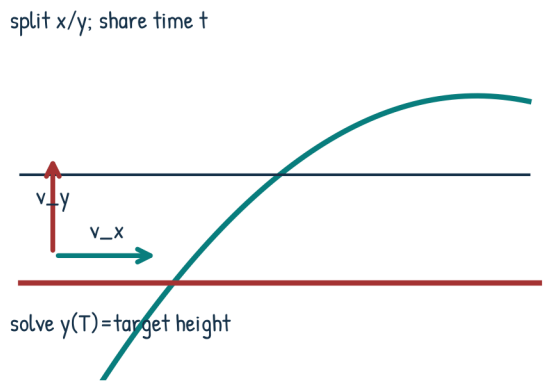
Impact speed and angle

At impact, compute vx and vy separately, then speed and direction. For a lower landing level, impact speed exceeds launch speed in vacuum.

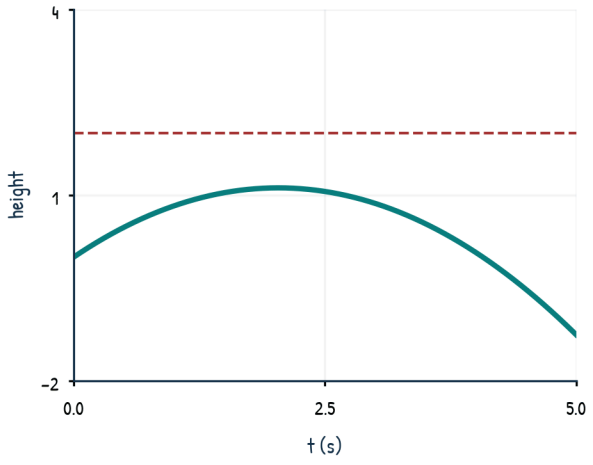
$$v = \sqrt{(v_0\cos\theta)^2 + (v_0\sin\theta - gT)^2}$$

The squared-speed change is $-2g(y_f - y_0)$; it is independent of launch angle.

elevated: physical sketch



elevated: diagnostic graph



REMEMBER / Pick the positive landing-time root, then multiply by horizontal velocity.

Projectiles on an inclined plane

Resolve gravity along and normal to the slope; define launch angle from the slope.

Use slope-aligned coordinates

Let the plane rise at angle alpha. Choose s along the plane uphill and n perpendicular outward. Gravity components are downhill along s and into the plane along n.

$$a_s = -g\sin\alpha, \qquad a_n = -g\cos\alpha$$

WHY / The launch angle beta is measured above the incline; the horizontal angle is alpha+beta.

Return to the same plane

Initial normal speed is v0 sinbeta. The normal coordinate returns to zero after a nonzero flight time. This time is finite for alpha below 90 degrees.

$$T = \frac{2v_0\sin\beta}{g\cos\alpha}$$

When alpha=0, recover ordinary ground-to-ground flight time for launch angle beta.

Range along the incline

Along-plane displacement combines initial along-slope speed v0 cosbeta with downhill acceleration. Substitute T from the normal motion.

$$R_s = \frac{2v_0^2\sin\beta\cos(\alpha+\beta)}{g\cos^2\alpha}$$

WATCH / This formula describes a return to the same uphill plane and a valid forward landing.

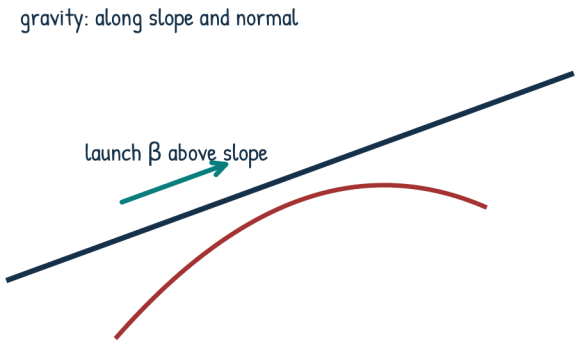
Optimize the slope range

At fixed launch speed and slope, maximize the product sinbeta cos(alpha+beta). The optimum measured from the horizontal is 45 degrees plus half the slope angle.

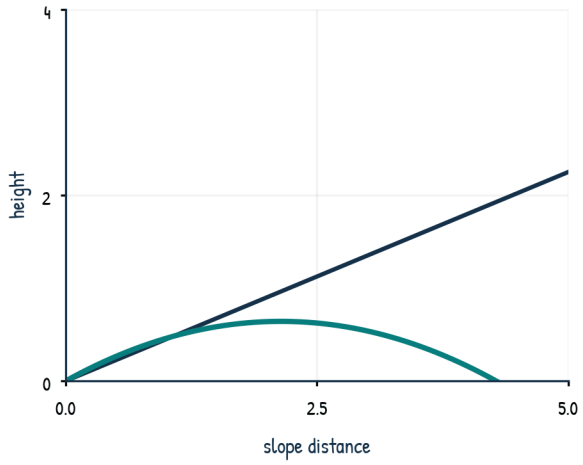
$$\beta_{\max} = 45^\circ - \frac{\alpha}{2}, \qquad \theta_{\text{horiz}} = 45^\circ + \frac{\alpha}{2}$$

At alpha=0, the best launch is 45 degrees above horizontal.

incline: physical sketch



incline: diagnostic graph



REMEMBER / For an uphill slope alpha and launch angle beta above it, T = 2v0 sinbeta/(g cosalpha).

Reachability and the safety envelope

The envelope gives the highest point attainable at each horizontal coordinate for fixed speed.

Family of trajectories

At fixed speed v0, each launch angle gives a different parabola. For a fixed x, maximize y over launch angle; the boundary is the envelope of the family.

$$y(x,\theta) = x\tan\theta - \frac{gx^2}{2v_0^2\cos^2\theta}$$

WHY / An envelope is not one projectile path; it is the boundary traced by the best angle at each x.

Find the maximum y at fixed x

Set the derivative with respect to launch angle to zero. The maximizing slope obeys tantheta=v0²/(gx), for forward x>0.

$$\tan\theta_* = \frac{v_0^2}{gx}$$

As x approaches zero, the optimum approaches a vertical launch.

Envelope equation and domain

Substitution gives the downward-opening safety parabola. For forward targets at or above launch level, a point below this boundary can be reached by some suitable angle before the projectile lands.

$$y_{\text{env}} = \frac{v_0^2}{2g} - \frac{gx^2}{2v_0^2}$$

WATCH / State the physical domain. The envelope does not make post-impact or underground portions of ideal parabolas physical.

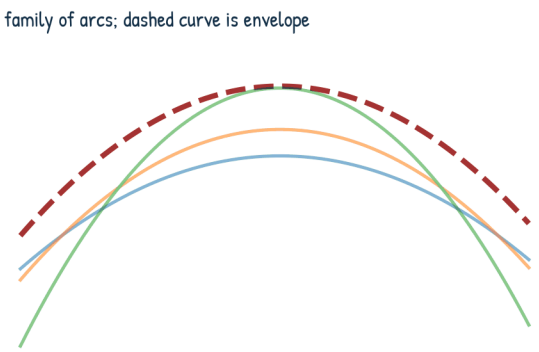
Read its intercepts

At x=0 the envelope height is the vertical-launch maximum. It meets y=0 at the maximum equal-height range. Dimensional and limit checks confirm the shape.

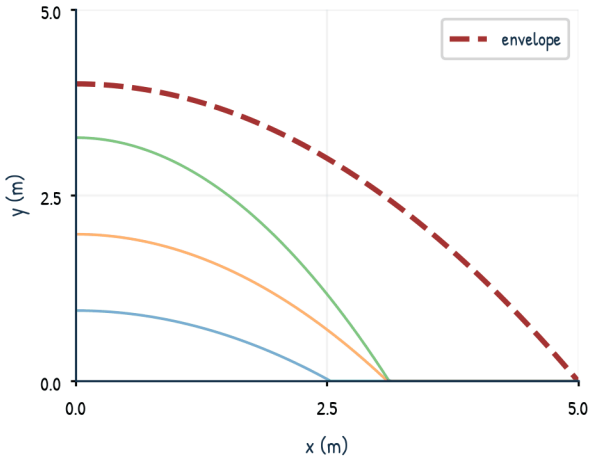
$$y_{\text{env}}(0) = \frac{v_0^2}{2g}, \qquad x_{\text{ground}} = \frac{v_0^2}{g}$$

Increasing v0 raises and widens the envelope; increasing g lowers and narrows it.

envelope: physical sketch



envelope: diagnostic graph



REMEMBER / For launch from the origin in vacuum, the envelope is y = v0²/(2g) - gx²/(2v0²).

Relative velocity in two dimensions

Write the frame relation as a vector equation before resolving components.

Subtract vectors, not speeds

All velocities must be measured in a compatible frame. The velocity of A relative to B is the vector difference; the magnitude is found only after subtraction.

$$\mathbf{v}_{A/B} = \mathbf{v}_A - \mathbf{v}_B$$

WATCH / Subtracting speed magnitudes loses direction and fails for non-collinear motion.

Use a velocity triangle

Draw v_B and v_A from a common origin; their difference is the vector from the tip of v_B to the tip of v_A. Label the reference frames on each velocity.

$$\mathbf{v}_{A/G} = \mathbf{v}_{A/B} + \mathbf{v}_{B/G}$$

WHY / Frame chaining is additive; reverse a relation by reversing the vector.

Rain and a moving observer

If rain falls vertically at speed vr and a person walks east at vm, rain relative to the person has a westward horizontal component. The apparent source lies forward of vertical.

$$\mathbf{v}_{r/m} = (-v_m,\ -v_r), \qquad \tan\alpha = \frac{v_m}{v_r}$$

If vm=3 and vr=4 m/s, the apparent direction is 36.9 degrees from vertical.

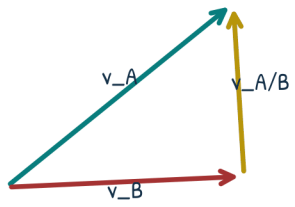
Rate of separation in a plane

For relative position vector r and relative velocity vrel, the instantaneous rate of change of separation is the radial projection of vrel, not generally its full magnitude.

$$\frac{d|\mathbf{r}|}{dt} = \hat{\mathbf{r}} \cdot \mathbf{v}_{rel} \qquad (|\mathbf{r}| > 0)$$

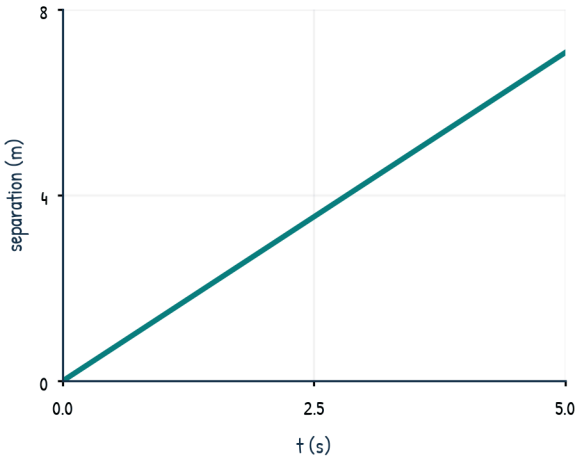
WATCH / At zero initial separation, use the right-hand growth rate, which is |vrel|; the radial unit vector is undefined exactly at coincidence.

relative2d: physical sketch



vector subtraction, not speed subtraction

relative2d: diagnostic graph



REMEMBER / v_A/B = v_A - v_B; the relative vector points along how A moves as seen by B.

River crossing: time, drift and aim

Separate across-river and along-river components of the boat relative to ground.

Define current and boat velocity

Let +x point straight across, +y downstream. Boat velocity relative water plus water velocity relative ground gives the ground velocity.

$$\mathbf{v}_{b/g} = \mathbf{v}_{b/w} + \mathbf{v}_{w/g}$$

If the boat points straight across, its across speed is vb and it drifts at current speed vr.

Point straight across

For river width d and boat speed vb relative water, straight-across heading gives crossing time d/vb and drift vr times that duration.

$$t = \frac{d}{v_b}, \qquad \Delta y = v_r \frac{d}{v_b}$$

WHY / This minimizes crossing time when the boat speed through water is fixed and the current is parallel to the bank.

Land directly opposite

Aim upstream so the upstream component cancels the current. This is possible only if boat speed is at least the current speed.

$$v_b \sin\alpha = v_r, \qquad t = \frac{d}{v_b \cos\alpha} = \frac{d}{\sqrt{v_b^2 - v_r^2}}$$

WATCH / The no-drift heading takes longer than pointing directly across. At vb=vr, crossing time diverges.

Aim at a specified landing point

For any chosen heading, compute ground components first. Time is width divided by across component; drift is downstream component times that time.

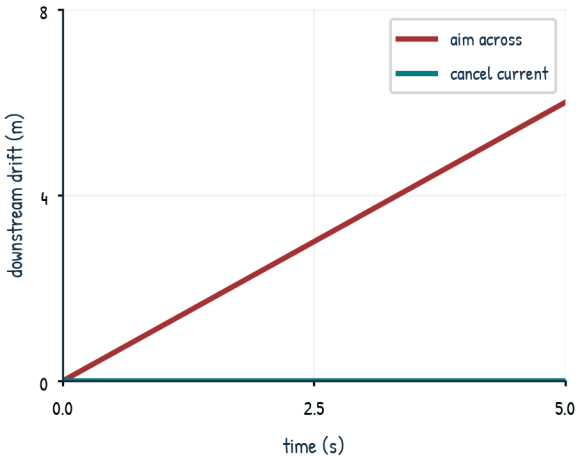
$$t = \frac{d}{v_{x,g}}, \qquad \Delta y = v_{y,g}\, t$$

Keep direction signs explicit: an upstream component is negative if downstream is positive.

river: physical sketch



river: diagnostic graph



REMEMBER / Crossing time depends on the across component; drift depends on the downstream component.

Aircraft and wind

The ground track is the vector sum of airspeed and wind velocity.

Start with the vector law

Airspeed is velocity relative to air; wind is air relative to ground. Their vector sum is the aircraft velocity over ground.

$$\mathbf{v}_{p/g} = \mathbf{v}_{p/a} + \mathbf{v}_{a/g}$$

WATCH / "Wind from west" means velocity toward east. Convert wind wording into a vector before choosing heading.

Crosswind correction

For a desired due-north ground track and an eastward wind wx, the plane must point west of north so its west component cancels wx. The north component then sets ground speed.

$$v_{p/a,\,x} = -w_x, \qquad v_{g,\,y} = \sqrt{v_a^2 - w_x^2} + w_y$$

A feasible heading requires |wx| no greater than the available airspeed.

Wind with along-track component

Resolve wind into along-track and cross-track parts. Cancel the cross-track component; add the along-track wind to the remaining aircraft component.

$$v_g = \sqrt{v_a^2 - w_\perp^2} + w_\parallel$$

WHY / A tailwind increases groundspeed; a headwind decreases it, while crosswind changes heading.

Sign-audited example

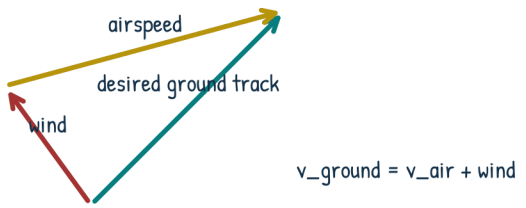
A plane needs 200 km/h north over ground. Wind from west blows east at 40 km/h. The air velocity needs a 40 km/h west component and 200 km/h north component.

$$v_a = \sqrt{200^2 + 40^2} = 203.96\ \text{km/h}$$

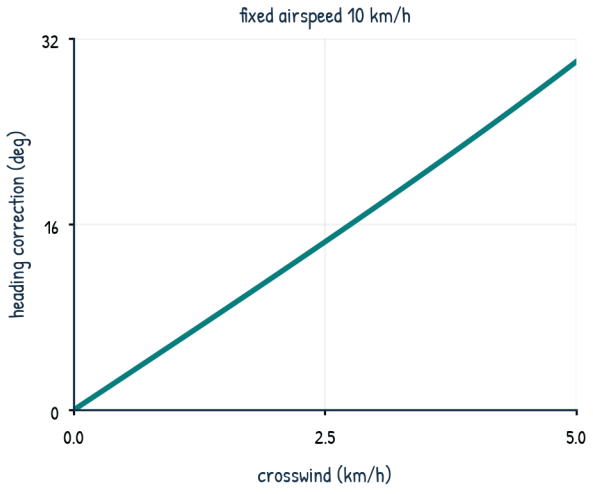
$$\alpha = \tan^{-1}(40/200) = 11.31\,^\circ$$

The heading is 11.31 degrees west of north, not east.

aircraft: physical sketch



aircraft: diagnostic graph



REMEMBER / Cancel crosswind with an opposite air-velocity component; compute ground speed from the along-track component.

Angular motion and tangential speed

Radians make arc length proportional to angle; angular rate converts to linear speed.

Angular position and rate

Measure angle in radians. Angular velocity is the rate of change of angular position; its sign states rotation sense under the chosen convention. After a reversal, total angular path accumulates |dtheta|, not signed endpoint change.

$$\omega = \frac{d\theta}{dt} \quad [\mathrm{rad/s}]$$

One revolution is 2π rad, not 360 radians.

Arc-length relation

For a circle of radius R, arc length is R times angle in radians. Differentiate to link tangential speed to angular speed.

$$s = R\theta, \qquad v_t = R\omega$$

WHY / Radians are dimensionless in SI, but retaining "rad" helps distinguish angular rate from ordinary frequency.

Period and frequency

Period is time per revolution; frequency is revolutions per second. For steady rotation, angular speed magnitude equals 2πf; for nonuniform rotation, 2π/T is the cycle-averaged angular speed.

$$|\omega| = \frac{2\pi}{T} = 2\pi f$$

WATCH / Do not use rpm directly in v=Romega; convert rpm to rad/s first.

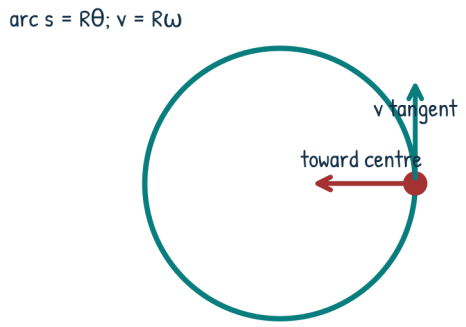
Quick conversion

For 120 rpm, divide by 60 to get 2 revolutions per second, then multiply by 2π. A rim of radius 0.5 m has tangential speed Romega and period 0.5 s. Its circumference is $2\pi r = \pi\ \mathrm{m}$, so two turns per second cover $2\pi\ \mathrm{m}$ each second.

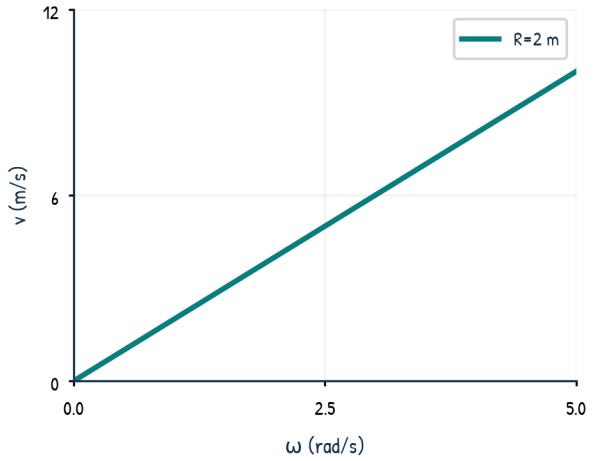
$$120\ \mathrm{rpm} = 4\pi\ \mathrm{rad/s}, \qquad v = 2\pi\ \mathrm{m/s}$$

Check units: m times s^-1 gives m/s.

circle: physical sketch



circle: diagnostic graph



REMEMBER / For fixed radius, v = Romega; one full turn is 2π radians.

Centripetal and tangential acceleration

Velocity turns even when speed is constant; changing speed adds a tangential component.

Differentiate the rotating unit vectors

On a circle, position is R rhat. The radial unit vector rotates at angular speed omega, and the tangential unit vector rotates toward -rhat.

$$\dot{\hat{r}} = \omega\hat{\theta}, \qquad \dot{\hat{\theta}} = -\omega\hat{r}$$

WHY / Acceleration contains a radial term because the direction of the velocity basis changes.

Full circular acceleration

Differentiate the velocity vector: radius times angular speed in the tangential direction. The radial term points inward; the tangential term changes speed. Its sign follows angular acceleration under the chosen positive rotation sense.

$$\mathbf{a} = -R\omega^2\hat{r} + R\dot{\omega}\hat{\theta}$$

At constant omega, only the inward component remains.

Magnitudes and meanings

Tangential acceleration is the time rate of speed change; centripetal acceleration changes direction. They are perpendicular, so the total magnitude is their root-sum-square.

$$a_t = \frac{dv}{dt} = R\dot{\omega}, \quad a_c = \frac{v^2}{R}, \quad a = \sqrt{a_t^2 + a_c^2}$$

WATCH / Centripetal acceleration is not a separate force; "centripetal force" is the net inward force in dynamics.

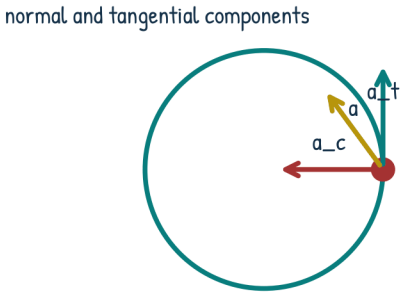
Quick computation

At R=2 m and omega=3 rad/s, inward acceleration is 18 m/s². If angular acceleration is 4 rad/s², tangential acceleration is 8 m/s².

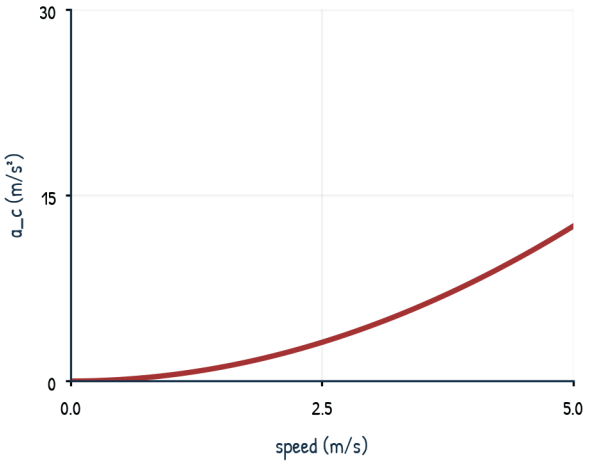
$$a = \sqrt{18^2 + 8^2} = \sqrt{388} \approx 19.7 \text{ m/s}^2$$

The acceleration vector points inward and along the tangent; it is not generally radial.

circ accel: physical sketch



circ accel: diagnostic graph



REMEMBER / a = -Romega² rhat + Ralpha thetahat; inward magnitude is v²/R.

Curvature and the normal acceleration

A curved path has an instantaneous osculating circle; normal acceleration sets its radius.

Resolve acceleration along velocity

At nonzero speed, decompose acceleration into a tangential component parallel to velocity and a normal component perpendicular to it.

$$a_t = \frac{dv}{dt}, \qquad a_\perp = |\mathbf{a} - (\mathbf{a}\cdot\hat{v})\hat{v}|$$

WHY / Only the perpendicular component bends the velocity direction.

Radius of curvature

Over a short time, velocity direction turns by dphi while arc distance is ds=v dt. Curvature is dphi/ds; the normal acceleration is v² times curvature.

$$\kappa = \frac{1}{\rho}, \qquad |a_\perp| = \frac{v^2}{\rho}$$

For a circle, rho is its ordinary radius and this becomes centripetal acceleration.

Projectile at launch and apex

For gravity downward, at launch angle theta the normal component magnitude is glcosthetal. At the apex, velocity is horizontal so all g is normal.

$$\rho_{\text{launch}} = \frac{v_0^2}{g|\cos\theta|}, \qquad \rho_{\text{apex}} = \frac{v_0^2\cos^2\theta}{g}$$

WATCH / For a vertical launch, the path is straight and rho is infinite; handle costheta=0 as a limit.

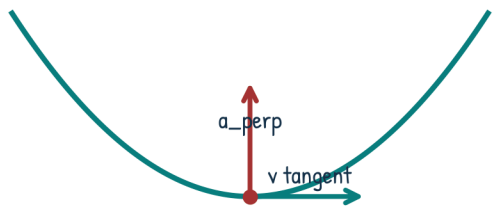
Limits and interpretation

If normal acceleration vanishes at nonzero speed, the instantaneous path is straight and curvature is zero. At a zero-speed cusp, the usual velocity-based decomposition may fail.

$$a_\perp \to 0 \Rightarrow \rho \to \infty$$

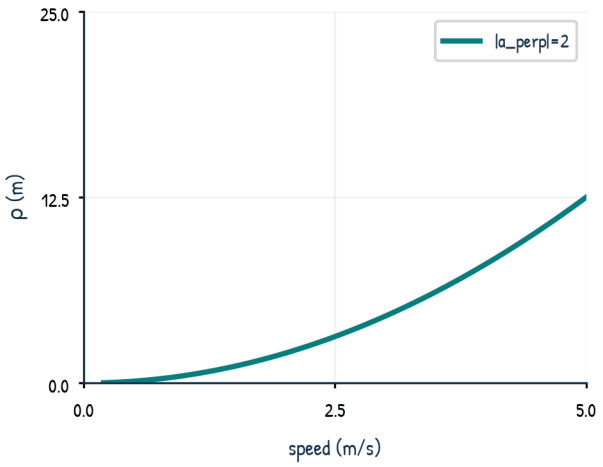
A smaller rho means tighter turning for the same speed.

curvature: physical sketch



ρ = v² / |a_perp|

curvature: diagnostic graph



REMEMBER / rho = v² / |a_perp|. Tangential acceleration changes speed; normal acceleration changes direction.

Kinematics method map and final checks

Start from what is given, choose a model, then test signs and limiting cases.

Route by the acceleration law

Constant a: use signed kinematic equations. Given a(t): integrate. Given a(v): separate dv/a(v). Given a(x): use v dv/dx. A graph may provide slopes or areas directly.

$$a = \text{constant} \rightarrow \text{SUVAT}; \quad a(x) \rightarrow v\,dv/dx$$

WHY / Use the simplest relation that matches the information; do not force every problem into one formula.

Route projectile questions

Split x and y and share time. Equal elevations: use T, H and R formulas. Unequal elevations: solve the vertical quadratic. Incline: resolve along and normal to the plane.

$$\text{projectile}: \; x(t), y(t); \quad \text{same } t$$

WATCH / Check the launch and landing levels before quoting the standard range.

Route relative and circular questions

Relative motion: write frame-chaining vectors and define separation. Circular motion: keep tangential and normal acceleration separate. General path: use local curvature.

$$\mathbf{v}_{A/B} = \mathbf{v}_A - \mathbf{v}_B; \quad a_c = v^2/R$$

Relative velocity is a vector; a_c is acceleration, not force.

Ten-second audit

Check units; sign convention; initial conditions; root time; distance versus displacement; model validity; and whether the path reaches the stated target before hitting another surface.

$$\text{limit checks}: \; a \rightarrow 0, \; g \rightarrow 0, \; v_r \rightarrow 0, \; \theta \rightarrow 0^\circ$$

If an answer changes under relabelling the positive axis, a sign error is likely.

model: physical sketch

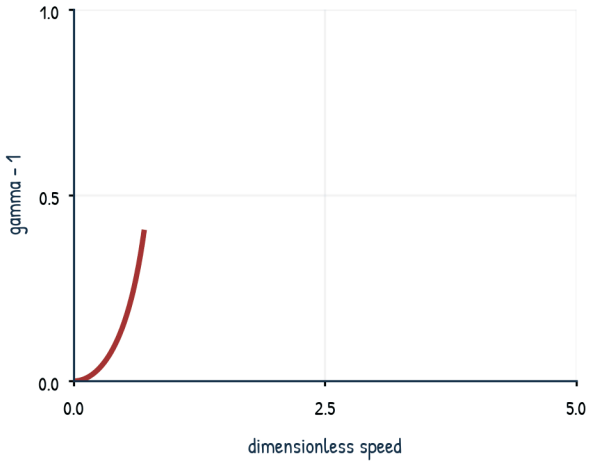
ideal model -> prediction



check regime, geometry, initial data

state what breaks the model

model: diagnostic graph



REMEMBER / Name the frame, axis, interval, acceleration law and requested quantity before calculating.